Lecture 9: Forward Rate Agreements

Questions

These five multiple-choice questions follow the lecture's reasoning with different numbers. Work through the problem before reading the solutions. Keep full precision until the final answer.

All rates are annual compounded semiannually. Use the lecture's end-date compounded payoff convention, not a discounted start-date FRA settlement.

1. Identify the exposure

A company receives $100,000 at t=1 and will invest it until t=3. It fixes a 6% annual rate today for that future investment period. Which move hurts its unhedged investment and should trigger a positive investor payoff?

A. The market rate for t=1 to t=3 rises above 6%

B. The market rate for t=1 to t=3 falls below 6%

C. Only a change in the amount invested matters

D. Today's one-year rate rises, regardless of the future two-year rate

2. Calculate the investment payoff

Use the $100,000 investment from t=1 to t=3, FRA 6% and realized market rate 4%. Rates are annual compounded semiannually. Under the lecture's end-date payoff convention, what does the investor receive from the derivative?

A. $12,550.88

B. $4,307.67

C. -$4,307.67

D. $4,000.00

Lecture 9: Forward Rate Agreements

3. Switch to a future borrower

A different company will borrow $100,000 from t=1 to t=2. It locks 4%; the realized matching market rate is 6%. Using semiannual compounding and an end-date payoff, what are its payoff and final interest cost?

A. +$6,090.00 payoff; $0.00 cost

B. +$2,000.00 payoff; $4,090.00 cost

C. -$2,050.00 payoff; $8,140.00 cost

D. +$2,050.00 payoff; $4,040.00 cost

4. Derive the implied forward rate

Today's one-year zero rate is 4% and two-year zero rate is 6%, annual with semiannual compounding. What forward annual rate applies from t=1 to t=2?

A. 8.0196%

B. 2.0000%

C. 5.0000%

D. 8.0000%

5. Interpret a negative payoff

Return to the $100,000 investment from t=1 to t=3 at a locked 6%. If the matching market rate is 8%, what happens to the final hedged interest?

A. It equals $12,000 because simple interest is used

B. It rises to $16,985.86; the hedge also gains

C. It remains $12,550.88; the investor pays on the derivative

D. It falls below zero because the derivative payoff is negative

Lecture 9: Forward Rate Agreements

Worked solutions

1. Identify the exposure - Answer B

Step 1. Locate the actual investment. The money arrives in one year and remains invested for two years.

Step 2. Identify the adverse move. A lower market rate means less interest earned on the investment.

Step 3. Choose the offset. The investor should receive on the derivative when the market rate is below the locked rate.

$$\text{Payoff}_3^{invest} = N[(1 + FRA_{1,3}/2)^4 - (1 + R_{1,3}/2)^4]$$

Interpretation. Use the market rate covering the same future start and end dates.

Check your reasoning. An investor and a borrower have opposite adverse rate moves.

2. Calculate the investment payoff - Answer B

Step 1. Count compounding periods. The two-year investment has four semesters. The one-year wait is not part of the investment period.

Step 2. Compute both terminal interest amounts. At 6%, interest is $12,550.881000; at 4%, it is $8,243.216000.

Step 3. Take locked minus market. The investor receives $4,307.665000, offsetting the smaller market interest.

$$100000(1.03^4 - 1.02^4) = 4307.665000$$

Interpretation. Market interest plus the derivative payoff equals the locked interest amount.

Check your reasoning. N x (FRA - R) x T is a simple-interest approximation, not our compounded calculation.

Lecture 9: Forward Rate Agreements

3. Switch to a future borrower - Answer D

Step 1. Compute market borrowing interest. 100,000 x (1.03^2 - 1) = $6,090.

Step 2. Compute the locked interest and payoff. Locked interest is 100,000 x (1.02^2 - 1) = $4,040. The borrower receives the $2,050 difference.

Step 3. Subtract the receipt from cost. Final interest cost is 6,090 - 2,050 = $4,040.

$$\text{Payoff}_2^{borrow} = 100000(1.03^2 - 1.02^2) = 2050$$

Interpretation. The borrower receives when market rates rise, which offsets the higher loan cost.

Check your reasoning. Do not add a positive derivative receipt to the borrower's cost.

4. Derive the implied forward rate - Answer A

Step 1. Compare two routes. Invest for two years today, or invest for one year and lock the second year.

Step 2. Divide growth factors. The second-year factor is 1.03^4 / 1.02^2. There are two semesters in the forward period.

Step 3. Solve for the annual quote. Take the square root, subtract one, then multiply by two: 8.019608%.

$$FRA_{1,2} = 2\left[\left(\frac{1.03^4}{1.02^2}\right)^{1/2} - 1\right]$$

Interpretation. The forward is implied by current prices; it is not a required prediction of future spot rates.

Check your reasoning. Subtracting spot rates or averaging them does not match the growth factors.

Lecture 9: Forward Rate Agreements

5. Interpret a negative payoff - Answer C

Step 1. Recognize the better underlying outcome. At 8%, the investment earns more interest than at the locked 6%.

Step 2. Calculate the offset. The investor payoff is 100,000 x (1.03^4 - 1.04^4) = $-4,434.98.

Step 3. Add underlying interest and payoff. The combination equals $12,550.88, independent of the realized market rate under matching terms.

$$N[(1+R/2)^4-1]+N[(1+FRA/2)^4-(1+R/2)^4]=N[(1+FRA/2)^4-1]$$

Interpretation. A negative payoff can be exactly what makes a hedge deliver its intended result.

Check your reasoning. Do not evaluate the derivative in isolation.